Piter Z. Garcia Bautista

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Research Profile

Graduate researcher and NSF Robert Noyce Scholar working at the intersection of quantum computing, compiler-level optimization, and reproducible experimentation. Current research spans quantum network optimization, fault-tolerant routing, and fairness-aware decision workflows. Specifically interested in compiler-assisted precision and resource management for fault-tolerant quantum workloads, as explored in the SPQR project and TAFFO-inspired frameworks for quantum error correction code selection.

Research Interests

Quantum compiler optimization and error correction code selection; fault-tolerant quantum circuit design; LLVM/MLIR-based compiler backends for quantum workloads; reproducible benchmarking and structured experimental design; quantum-classical hybrid methods.

Education

Rochester Institute of Technology — Expected Aug 2026
Master of Science, Data Science — GPA 3.9 — *Rochester, NY*

- Concentration: Bioinformatics & Artificial Intelligence.
- Relevant coursework: Foundations of Data Science (DSCI-633), Software Engineering for Data Science (DSCI-644), Applied Data Science I (DSCI-601), Bioinformatics Algorithms (BIOL-630), Data-Driven Knowledge Discovery (ISTE-780).

University of Rochester, Warner School of Education — Expected Aug 2026
Master of Science track, Teaching Computer Science K–12 — GPA 4.0 — *Rochester, NY*

- NSF Robert Noyce Teacher Scholarship Program ($30,000 full funding).
- Advanced Certificate in Leadership in Disability and Inclusive Practices.

Rochester Institute of Technology — 2015
Master of Science, Computer Science — GPA 3.2 — *Rochester, NY*

- Concentrations: Computer Graphics, Artificial Intelligence, Big Data Analytics.
- Coursework in compiler design, algorithm analysis, and distributed systems.

Farmingdale State College — 2012
Bachelor of Science, Computer Engineering Technology — GPA 3.3 — *Farmingdale, NY*

- Dual degree: Electrical Engineering & Computer Engineering Technology.
- Honors: CSTEP Award & Merit, Dual BS Scholarship.

Research and Professional Experience

Rochester Institute of Technology Fall 2025 – Present
Graduate Assistant, Quantum and AI Research *Rochester, NY*

- Conduct research on quantum network path optimization and fault-tolerant routing under adversarial and stochastic conditions.
- Develop compiler-level experimental infrastructure for quantum circuit workloads using Qiskit and Cirq.
- Implement and validate quantum error correction code selection workflows in simulation (Stim-compatible pipelines).
- Apply LLVM/MLIR IR-level analysis concepts to quantum compilation pipeline studies; reviewed TAFFO precision-tuning methodology for applicability to quantum resource management.

University of Rochester, Warner School of Education Sep 2024 – Present
CS Teacher Candidate & Education Research (NSF Noyce Scholar) *Rochester, NY*

- Deliver K–12 computer science instruction in active public school placement as part of the NSF Robert Noyce Scholar program, including computational thinking, inclusive pedagogy, and NYS learning standards.
- Conduct applied education research on AI-informed and data-driven approaches to improve learning outcomes for neurodivergent students, applying Universal Design for Learning (UDL) and evidence-based adaptive strategies.
- Produce lesson artifacts and instructional analyses documenting the intersection of AI, inclusive education, and evidence-based CS pedagogy.

VEDADATA Inc. Sep 2022 – Aug 2024
Data Solutions Engineer *New York, NY*

- Designed scalable Python workflows for ingestion, validation, and statistical diagnostics, emphasizing data reliability and structured quality checks.
- Strengthened production reliability through anomaly detection, schema validation, and structured logging.

VIOME Inc. Jan 2021 – Sep 2022
AI Data Solutions Engineer *New York, NY*

- Developed healthcare ML workflows for feature engineering, model experimentation, and reliability-aware evaluation in AWS-based production environments.
- Implemented uncertainty-quantification and model-monitoring components to flag prediction instability under distribution shift.

Selected Technical Writing

- *Quantum Network Path Optimization and Fault-Tolerant Routing* (manuscript under review, 2025–2026) — quantum-classical hybrid ML for fault-tolerant routing under noise constraints and decoherence.
- *Big Data Medical Diagnosis: A Multi-Method Approach* (RIT M.S. Computer Science graduate research, 2015) — ensemble and deep learning methods for clinical classification on high-dimensional biomedical data.

- *Predicting Hospital Readmission Rates: A Multi-Method ML Analysis* (DSCI-633 Project Report, 2025) — ensemble and regularized regression methods for clinical risk stratification under distribution shift.
- *BIO614-FinalProjectProposal* (Overleaf manuscript, 2026) — RNA secondary structure prediction with thermodynamic deep learning enhancements, reproducible benchmarking, and biological interpretation.
- *Equitable Bioinformatics: Enhancing Diagnostic Decision-Making through RNA and Biomarker Data* (ISTE-780 Project Report, 2025) — fairness-aware ML for genomic diagnostic algorithms with SHAP-based bias detection and statistical significance testing.
- *Differential Gene Expression in Murine DRG Neurons Following Sciatic Nerve Injury: An NGS Reanalysis* (BIOL550 Computational Biology Project, 2026) — reproducible bulk RNA-seq pipeline using DESeq2, QC validation, and pathway-level biological interpretation of nociceptor gene expression.
- *Fairness-Aware Bandits for Clinical Decision Systems* (DSCI-601 Project Proposal, 2025) — bandit learning with equity constraints applied to high-stakes clinical diagnostic decision-making, addressing demographic disparity in sequential clinical recommendation workflows.

Awards & Recognition

- NSF Robert Noyce Teacher Scholarship (2024–2026) — full funding for M.S. in Teaching Computer Science K–12.
- MS International Scholarship (2012–2015) — MESCYT & RIT, awarded for academic excellence.
- Honorable Mention Technology Award (2012) — NYS STEP/CSTEP, 2nd place capstone project.
- IEEE Alumni Member (2009–Present) — Member No. 90613000.

Technical Competencies

Programming: Python, R, Java, C++, C#, MATLAB, SQL, JavaScript, Bash
Quantum Computing: Qiskit, Cirq (gate-based quantum circuit simulation and algorithm development)
Compiler & Systems: LLVM/MLIR compiler backend framework; TAFFO/Precimonious (approximate computing and precision tuning)
Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, fairness-aware ML, contextual bandit workflows
Data & Analysis: pandas, NumPy, SciPy, SHAP analysis, statistical testing, reproducible benchmarking
Infrastructure: AWS, GCP, Docker, Git/GitHub, Linux, LaTeX

Selected Fit for NTNU Quantum Compiler Technologies Position

- Directly relevant research in quantum circuit optimization and fault-tolerant routing; active study of compiler-assisted precision tuning methodologies (TAFFO, Precimonious) and their extension to QEC code selection.
- Hands-on experience with LLVM/MLIR IR-level analysis and quantum circuit frameworks (Qiskit, Cirq) providing a practical foundation for compiler backend contributions within the SPQR project.

- Reproducibility-first experimental discipline (multi-environment: local, Colab, GCP) with traceable infrastructure—well matched to SSG's open-source and benchmark-driven research culture (CGO, PLDI).
- Interest in PoliMi collaboration; familiar with TAFFO's Value Range Analysis approach as the classical precedent the SPQR project extends to the quantum domain.